**Pokhara University**
**Faculty of Management Studies**

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| Course Code: CMP 281 | Full marks: 100 |
| Course Title: **Project II** | Pass marks: 45 |
| Nature of the course: Theory + Practical | Credit Hrs: 3 |
| Year: Second, | Total periods: 48 hours |
| Level: Bachelor | Program: BCSIT |
| Semester: IV | |

## 1. Course Description

This comprehensive full-stack Web Development course is designed to empower students with a deep understanding of both front-end and back-end technologies, utilizing diverse frameworks. Through hands-on experience, participants learn to construct dynamic and responsive web applications while exploring essential concepts, tools, and contemporary development practices. Additionally, the hands-on project requires students to build a computer application, encouraging them to develop software for different devices using programming languages they are already familiar. This project involves fundamental operations, including data creation, retrieval, update, and deletion, with the option to incorporate advanced algorithms. While leveraging languages learned up to the 4th semester is recommended, students are encouraged to explore additional technologies, supported by the use of Computer-Aided Software Engineering (CASE) tools for efficient development. The project themes encompass information systems, e-commerce portals, and game applications, emphasizing independent code creation and minimizing reliance on pre-existing components unless indispensable.

## 2. Course Objective:

- Develop leadership skills for effective guidance and oversight in software project development.
- Cultivate teamwork abilities, emphasizing collaboration and communication within project teams.
- Acquire proficiency in using Computer-Aided Software Engineering (CASE) tools to streamline and enhance software development processes.
- Develop a solid comprehension of both front-end and back-end web development principles, covering key concepts and methodologies,
- Attain expertise in a modern front-end framework (e.g., React, Angular, or Vue) for crafting interactive user interfaces, and in a back-end framework (e.g., Express.js, Django, or Flask) for server-side development.
- Enhance programming skills by writing and optimizing code, gaining practical experience in software development.

## 3. Focus of the Study

- Every student within the group is expected to actively participate in identifying and understanding the problem to be addressed by the project.
- System Analysis : Feasibility Study, System Requirement Specification (SRS)

- System Design: Architecture Design, Interface Design, Database/Procedure/Algorithm Design
- Implementation and Testing

**4. Evaluation system and Student's Responsibility**

| SN | Evaluation Details | Day for Evaluation (48 hrs.) | Marks |
|---|---|---|---|
| 1. | **Introduction to project I** | Week -1, Project Seminar (Introduction) | 5 marks for attendance and submission of topic on time |
| 2. | **Topic finalization** | Week -2(Discussion on project topic, Research) | |
| 3. | **Proposal Defense** | Week-3(Preparation for proposal defense) | 10 |
| 4. | **Mid- Term Defense** | Week-10 (Presentation, printed copy of project mid-term report, Demo,) | 15 |
| 5. | **End- Term Defense** | Week 12 (Presentation, Demo 80% of the work is to be completed) | 25 |
| 6. | **Final Defense** | Week 14 (Final Project presentation, Demo and Q&A) | 45 |
| | | **Total** | 100 |

**5. Project Work Phase:**

The entire project work is divided into three phases and evaluation will be done accordingly.

<table>
<tr><th>Phase</th><th>Description</th></tr>
<tr><td>Phase 1: Conceptual Framework and Proposal</td><td>Team formation: Students form teams of 2-4 members based on their interests, skills, and the nature of the project.<br>Conceptual Framework: Teams develop a conceptual framework for their project. This should include:
<ul>
<li>Problem statement: Clearly define the problem the project aims to address.</li>
<li>Objectives: Outline the goals and expected outcomes.</li>
<li>Scope: Define the boundaries and limitations of the project.</li>
<li>Significance: Explain the relevance and importance of the project.</li>
<li>Methodology: Briefly describe the approach and methods to be used.</li>
</ul>
Documentation<br>Teams document the conceptual framework in the format provided by the department for project proposals.<br>Presentations<br>Teams present their proposal in front of the examiner, providing a clear overview of their project</td></tr>
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| **Phase2: Progress Report** | 1. **Progress Report**<br>Teams demonstrate the progress made in the design phase, including.<br>▪ **Overall system design:** Present the architecture of the system<br>▪ **Architectural design:** Detail the structure and components of the system<br>▪ **Validation scheme:** Explain how the system's functionality will be validated<br>**2. Documentation of Progress:**<br>Teams document the progress made, highlighting key design decisions and any challenges encountered.<br>**3. Presentation to Internal Evaluator committee:**<br>Teams present their progress report to the internal evaluator committee for feedback and assessment. |
| **Phase3: Final Presentation and Defense** | 1. **Completion of all Phase:** Ensure that all phases of the project, including design, development, and testing, are completed.<br>2. **Product Output: Teams** develop the final output of their product using web technology. This could be a web application, platform, or any relevant web-based solution.<br>3. **Oral Defense:**<br>▪ Teams present the final product and its features to the external examiner.<br>▪ Conduct an oral defense, addressing questions from the examiner regarding design choices, implementation, and overall project execution<br>4. **Use web Technology / Java** : Emphasize the incorporation of web technology in the project. This could include technologies like HTML, CSS, JavaScript, and relevant frameworks for web development<br>5. **Submission of final Documentation:** Provide comprehensive documentation covering all aspects of the project, from the initial proposal to the final product. |

By following these three phases, students can systematically plan, execute, and present their project work, with a focus on utilizing web technology for effective solutions

**6. Project Evaluation Criteria**

**The marks of Mid-term,End- term and Final Defense are distributed as:**

- Report quality 10 %
- Presentation 5%
- Demonstration 15%
- Q&A 15%
- Teamwork 5%

**Note:**

There can be three members in final defense and the marks are adjusted in average basis: (Project supervisor, Program Coordinator and External)

**The project document shall include the following items:**

- Project Team members (At least 2 and maximum of 4)
- Project supervisor(s)
- Technical description of the project.
- Project Analysis, Design, Coding, Testing and Implementation details

**7. Students' Requirements**

Each student must secure at least 45% marks separately in both internal assessment and practical evaluation with a minimum of 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are required to attend all the evaluation criteria and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

**Report Contents:**

**1. Prescribed content flow for the project proposal**

1. Introduction
2. Problem Statement
3. Objectives of Study
4. Methodology
    a. Requirement Identification
    - Study of existing system
    - Requirement Collection

    b. Feasibility Study
    - Technical
    - Operational
    - Economic

    c. High Level Design of System (system flow chart/ methodology of the proposed system/ working mechanism of proposed system)
5. Gantt Chart (showing the project timeline)
6. Expected Outcome
7. References

**2.Prescribed content flow for the project report**

1. Cover & Title Page
2. Certificate Page
    - Supervisor Recommendation
    - Internal and External Examiners' Approval Letter
3. Abstract Page
4. Acknowledgements
5. Table of Contents
6. Abbreviations
7. List of Figures
8. List of Tables

7\. Main Report

8\. Appendices (Screen Shots/ Source Codes/ Supervisor Visit Log Sheets)

9\. References

**3. Prescribed Chapters in Main Report**

**Chapter 1: Introduction**

1.1. Background of the Study
1.2. Statement of the Problem
1.3. Objectives of the Study
1.4. Scope of the Study
1.5. Limitations of the Study

**Chapter 2: Review of Literature**

2.1. Description of fundamental theories, general concepts and terminologies related to the project
2.2. Review of the similar Projects and Theories

**Chapter 3: System Analysis and Design**

3.1. System Analysis

3.1.1. Requirement Analysis

- Functional Requirements (Illustrated using use case diagram/list)
- Non-Functional Requirements

3.1.2. Feasibility Analysis

- Technical
- Operational
- Financial
- Schedule

3.1.3. Data Modeling

3.1.4. Process Modeling

3.2. System Design

3.2.1. Architectural Design

3.2.2. Database Schema Design

3.2.3. Interface Design (UI Interface / Interface Structure Diagrams)

3.2.4. Physical Model

**Chapter 4: Implementation and Testing**

4.1. Implementation

4.1.1. Tools Used (CASE tools, Programming languages, Database platforms)

4.1.2. Implementation Details of Modules (Description of procedures/functions)

4.2. Testing

4.2.1. Test Cases for Unit Testing

4.2.2. Test Cases for System Testing

**Chapter 5: Conclusion and Future Recommendations**

5.1. Lesson Learnt

5.2. Conclusion

5.3. Recommendations

**Appendices (Screen Shots/ Source Codes/ Supervisor Visit Log Sheets)**

**Citation and Referencing**

The listing of references should be listed in the references section. The references contain the list of articles, books, that are cited in the document. The books, articles, and others that are studied during the study but are not cited in the document can be listed in the bibliography section.

**Report Format Standards**

**1.Page Number**

The pages from certificate page to the list of tables/figures should be numbered in roman type starting from i. The pages from chapter 1 onwards should be numbered in numeric starting from 1. The page number should be inserted at bottom, aligned center.

**2. Page Size and Margin**

The paper size must be a page size corresponding to A4. The margins must be set as 6 Top = 1; Bottom = 1; Right = 1; Left 1.5

**3. Paragraph Style**

All paragraphs must be justified and have spacing of 1.5.

**4. Text Font of Document**

- The contents in the document should be in Times New Roman font
- The font size in the paragraphs of document should be 12 E. Section Headings

- Font size for the headings should be 16 for chapter title, 14 for section headings, 12 for the sub-section headings. All the headings should be bold faced.

**5. Figures and Tables**

Position of figures and tables should be aligned center. The figure caption should be centered below the figure and table captions should be centered above the table. All the captions should be of bold face with 12 font size.